

MP305B firmware recovery and independent comparison

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MP305B V51, 2026-09-30. Research revision: uncommitted.

1 Outcome

The recoverable host-interface work in `~/mp305b-fw-re` is complete. All 33 probe jobs finish with checked emulator returns. There are 657 asserted cases: 305 encoder comparisons, 300 decoder comparisons, eight bridge-route scenarios and 44 targeted recovery cases. The 324 command-table rows are recorded observations and are counted separately.

The recovered bundle contains 2453 main, 255 CH58x and 380 PD function exports, plus 322 named locations and preserved aliases. Different function entry sets prevent treating these totals as coverage or adding them to the existing canonical exports. Both analyses remain available.

The original binary inputs, projects and notes are preserved. No hardware commands, commits, production implementation or V-model gate approval are part of this work. Complete sources, results, logs and regeneration commands are linked from the recovery record.

2 Evidence and independence

Instruction inspection, decompiled code and original-instruction execution are **confirmed in code** evidence. Emulator checks use synthetic memory, modeled peripherals and selected external-call substitutes. **Inferred** meanings retain that label. Existing capture comparisons are evidence from older recorded hardware sessions; no new hardware confirmation is claimed.

The recovered notes began from independent Ghidra imports. Their comparison sections also use the previous research and the existing WCH API map. They are not independent rediscoveries of every SDK symbol or hardware role. A firmware-to-firmware comparison is distinct from the project's use of "cross-checked" for independent derivation from WebLink JavaScript.

The statement tables in the appendices compare the source documents as seen during the original independent pass. Earlier files have since been corrected. Recovery corrections below take precedence over the historical verdict wording. Full vectors and implementation notes are in the linked Markdown files rather than compressed into the printed tables.

3 Corrections verified during recovery

Subject	Reviewed result	Evidence
Startup VTOR	SystemInit writes 0x10000 at 0x1DB8C	Instructions and executed write
Main decoder	State +0x208, fill index +0x20C, escape count +0x213	Instructions and stepped execution; rejects the independent note's proposed offset correction
CH58x global pointer	0x20002000; AUIPC is relative to its own PC	First two startup instructions executed
Reset path	Mailbox 0x2005F000 = 0x1234, then AIRCR reset request	Original instruction write trace; reset helpers marked non-returning
AF01 prefix	First incoming byte discarded; outgoing 31 is a route marker	Bridge probes and asserted byte traces
Waveform flag	Timer absent: 1; timer present: S+0x11	Four reply-builder cases
Fault addresses	Source 0x1FFFAA1E, published copy 0x1FFFB6A	Producer and publisher inspection
Fault meanings	Software OCP/OVP paths, specific thresholds and hysteresis	Producer inspection and 12 boundary cases

Subject	Reviewed result	Evidence
Permission scope	Four control-mode cases reject absent grant; BE accessory input reaches an enable request separately	Dispatcher and worker execution with a recorded actuator call
USB reassembly	127 stream bytes split into 62, 62 and 3; two frames in one OUT report lose the first in the tested path	Original bridge instructions

The BE result describes a command reaching the output request. It does not establish physical activation, electrical behavior, or the response of a real unit after its user presses Deny. The project continues to disconnect after denial under UR-008.

4 Completed jobs and remaining work

Job	Completion evidence
Recover source material	SHA-256 snapshot of 110 original workspace files; original notes and projects retained
Compare input images	Companion slices and initialized RAM match; main differs from restored image only in WebLink's four identity-magic bytes
Main command analysis	12 probe groups, 324 regenerated table rows, reviewed command and state notes
Host link	12 probe groups, including the 605 asserted framing comparisons
CH58x bridge	Eight diagnostic groups and eight asserted scenarios in the routing job
Additional verification	44 asserted cases with actual and expected results
Ghidra completion	Pending main and bridge helpers, deterministic naming, three corrected reset attributes, all requested decompilations complete
PD baseline	Existing 380-function export recovered; canonical 1007-entry table recovery retained
Register-map comparison	Seven vendor SVDs rescored; exact main MCU still unresolved
Report	Markdown, LaTeX and inspected PDF

The import of the pending main helper adds 143 candidate entries; the bridge helper adds 12 callbacks. These counts do not establish that every entry is a distinct source function. Main register labels retain the original HC32F4A0 SVD assumptions. Multiple HC32 variants match similar addresses; the clock-controller conflict at 0x40054000 remains unresolved.

The original host-interface scope did not include complete PD or power-stage internals, bootloader execution, physical pin tracing or replacement firmware. Those remain on the repository research list. The recovered update vectors use a synthetic flash array; they are not a validated device-update recipe.

5 Coverage of the repository research documents

Document	Recovery treatment
firmware.md	Detailed comparison in appendices; VTOR, D2 layout, companion identity and reset text corrected
protocol.md	Detailed comparison in appendices; route marker, waveform field and fault descriptions corrected
firmware-architecture.md	Host, bridge, startup and transport sections compared; RTOS conclusions retained
firmware-command-map.md	Dispatcher routes and handlers supplemented by the complete command notes
firmware-permissions.md	Shared control checks and accessory-input path added; policy retained
firmware-readable.md	Existing readable-C reconstruction retained; recovered named decompilation is an additional view
firmware-rtos.md	Existing scheduling/resource reconstruction retained; no new timing, stack or heap-lifetime measurement
firmware-verification.md	Earlier 2835-case run and manifest retained as historical evidence
hardware.md	Bus-role evidence retained; recovered labels do not identify exact silicon
hardware-buses.md	Main UART routing and CH58x endpoints compared and expanded in host-link notes
hardware-registers.md	Existing three-image access inventory retained; no claim to resolve every indirect access
hardware-ui-and-analog.md	Fault thresholds and local waveform semantics supplemented; physical topology remains inferred
device.md	Manufacturer specifications and rear-port distinction retained; no new unit inspection
README.md and captures	New bundle indexed; all four immutable capture hashes checked

For documents outside host interoperability this is an explicit scope review, not a claim that every statement was independently re-proved. Canonical per-function artifacts were checked for preservation, rather than represented as thousands of newly analyzed documents.

6 How to read the appendices

C or Confirmed means agreement with the cited code evidence. X or Corrected identifies a corrected earlier statement. N or New identifies an addition relative to the source snapshot. Inferred interpretations and unresolved hardware questions remain visible. The annotations refer to the original analysis methods described in the full command, host-link and CH58x notes.

7 Command and telemetry comparison

Full source: commands analysis.

Scope: `firmware.md` from “Command dispatcher” through “Block write and the 0xF0 to 0xFE commands”, and `protocol.md` sections 3 and 4. Addresses in `firmware.md` are file offsets; +0x10000 gives the processor addresses used here. “Confirmed” means the statement was re-derived from the disassembly or by execution in this pass.

7.1 firmware.md, “Command dispatcher”

Statement	Verdict
Function at 0x2F34, watches four flag bytes, first set flag selects a record in 0x1FFF9660, stride 260	Confirmed (0x12F34, flags 0x1FFE01AC..AF)
Byte 0 is the type byte, byte 2 the length, byte 4 the opcode	Confirmed. New: byte 0 is the high nibble and byte 1 the low nibble of the frame’s address byte; type 1 = USB host, 3 = CH58x system, 5 = CH58x BLE-central side, 6 = BLE host (section 1)
A request the switch does not recognise falls through without a reply	Confirmed, with the full list of ignored opcodes (2.1)
Most handlers write <code>opcode + 1</code> ; 0x20 stores 0x20	Confirmed
Type 6 appends one extra byte copied from the end of the request	Confirmed; it is the BLE route byte (31 AF01, 00 AF02) added by the CH58x
F2, F4, F6, FC, 20 append the constant 0x31	Confirmed
Type 6 is not by itself a transport name; BD treats types 5 and 6 differently	Corrected: type 6 is the BLE host route (CH58x ram:0x4074 addresses every GATT write 06 02); type 5 is the CH58x’s own BLE-central side
Shape (request, length, reply, type) for A2 C2 C4 C6 C8 D0 D2 D4 D6 DA DC DE E1 E2 E4 E8 EA EC EE	Confirmed at every call site
Shape (request, reply, type, length) for 00 20 E0 F0 F2 F4 F6 FC FE	Confirmed (F2, F4, F6 ignore the length)
BB, BD, BE get the length in r1 and no reply buffer	Confirmed; BD also gets the type in r2
Payload byte N is request byte N+1	Confirmed
E2 and E8 each have a block the decompiler dropped	Confirmed; both blocks are written out from the disassembly in 5.18 and 5.20 and executed

7.2 “Opcodes with their own compare”

Row	Verdict
00 0xDFFC, 01 91 67 01	Confirmed
18 inline 0x3126, sets S+0x47, stores the type, no reply	Confirmed (type goes to K+6)
20 0x16A8 block write, reply opcode 20	Confirmed
51 inline 0x312E, type 3 and payload 0, clears a flag, stores 1000	Confirmed; new: it acknowledges main’s 50 link-state message (2.3)
53 same shape, the other flag	Confirmed; acknowledges main’s 52
A0 inline 0x307E	Confirmed
A2 0x8368	Confirmed
BB 0x5AA0 list ingest	Confirmed; it is the CH58x’s BLE scan list
BD 0x4660 connection state and peer metadata	Confirmed; type 6 sets the host link S+3
BE 0x855C	Confirmed
C2 C4 C6 C8 D0 D2 D4 D6 DA handlers	Confirmed
D8 inline 0x31F0	Confirmed

7.3 “Opcodes 0xDC to 0xFE”

Statement	Verdict
Table branch at 0x304E covers DC..FE	Confirmed (TBB at 0x1304E, 35 entries at 0x13052)
The response opcodes in that range go to the no-reply path	Confirmed, plus E6, E7, F8..FB
DC 0x565C two payload bytes	Confirmed: id and step count of the selected slot
DE 0x5694 69 bytes	Confirmed
E0 0xB634 two layouts	Confirmed; layout chosen by route (6 = BLE short, else long)
E1 0x3C40 acts only for type 3, no reply	Confirmed; it stores the CH58x’s version
E2 0xB4D0, E4 0x5630, E8 0xD520, EA 0x54AC, EC 0x555C, EE 0x3D44	Confirmed
F0 replies F1 00 only for payload AC	Confirmed
F1 inline 0x30D8, type 3 and payload 0, no reply	Confirmed
F2 F3 00 + status	Confirmed
F4 F5 00, four echoed bytes, status	Confirmed (the four bytes are the SPI address)
F6 F7 00 + status	Confirmed
FC log slot only for payload CA	Confirmed; the slot holds the update request code 1/3/7
FD inline 0x32C0, payload 3 or FF, no reply	Confirmed
FE 0xDB94, FF AA 55 or FF 00 00	Confirmed
WebLink does not name 00 20 51 53 A0 BB BE F0..FE; they are in the image	Confirmed (not re-checked against WebLink)

7.4 “0xC2 reply, handler 0x58BC”

Statement	Verdict
Always writes through payload 35 and returns 37 (38-byte BLE frame with address)	Confirmed (emu)
Field table payload 0..32 and sources	Confirmed byte for byte (emu)
Payload 31: 1 if a global word is 0, otherwise S+17	Confirmed; the global is the device waveform-page timer 0x1FFE02A0
Payload 32: offset 76 of a second global	Confirmed: 0x1FFFA934 + 0x4C = 0x1FFFA980, raw output-on time in ms
Little-endian	Confirmed
“The scale of each field is not in this copy loop”	Resolved: section 7.1 gives the producer and unit of every field

7.5 “0xC4 reply, handler 0x5EF8”

Confirmed entirely (12 bytes, table, `usbLine` always present). New: the same builder is also sent unsolicited when `FLAGS` bit 12 is set.

7.6 “0xC8 command, handler 0xB7F4”

Statement	Verdict
r5[0] is the opcode; reply opcode + status; 0 applied, 1 remote not granted, FF rejected; length 2 or 3	Confirmed (emu)
Mode not 0: FF, the common epilogue still runs, an existing request value 0 clears the grant	Confirmed. New: rc = 0 then does not release the grant (the request byte is not written), so release must use the mode's own command
rc above 2: FF	Confirmed
rc 1 without grant: status 1, nothing applied	Confirmed
rc 0 clears S+66	Confirmed (through the epilogue)
rc 2: if S+3 == 2 both set to 1, otherwise length 0	Confirmed. New: S+3 == 2 means a USB host; over BLE the request waits for the front-panel prompt and is answered later by a deferred C9 00/C9 01 (section 3)
Different requested mode: output off, transfer state not 1, target below 4, stores S+0x31, skips the other checks	Confirmed (emu with FFFF setpoints)
Writes are sequential; a valid voltage can be stored before a bad current	Confirmed (emu #8)
Field table (3050, 5100, realChange 3 -> S+24, voltageSlow 1 -> S+16, currentOver 1 -> S+18, output via 0xA128/0xA134/0xAEBC/0xCB8C, model 3, refresh -> 0xA5FC)	Confirmed. Corrections: model is only compared, 0xA128 is not “called for a different model” but used as the output-off guard; 0xCB8C is the buzzer, not output control
3050 = 30.50 V, 5100 = 5.100 A	Confirmed by the units of the targets 0x1FFFA940 (10 mV) and 0x1FFFA944 (1 mA); new: an output voltage of 33.001 V or more for 500 ms sets fault bit 6
Bit 2 of the word at +264 skips the setpoint update and still builds a reply	Confirmed; new meaning: it is the deferred-reply mode set by the remote prompt callbacks, used when the transmit service calls the handler with C8 31

7.7 “State blocks”

Row	Verdict
S = 0x1FFFAACC, the dispatcher's fp	Confirmed
0x1FFFA354 program table, ten slots	Confirmed; layout in 6.3
0x1FFFA138 PD profile table	Confirmed; layout in 6.4
0x1FFFA340 PD class byte per index	Confirmed; it is the profile wattage
0x1FFFA8C4 program working bytes	Corrected: storage-worker request flags (+0 PD save, +1 last D2 index, +2 header save, +3 op, +4 steps save, +5 step-save clears busy, +6 load for D8, +7 slot, +8 DA cursor, +9 table re-save check)
0x1FFF8F7C step records, stride 12	Confirmed; transfer buffer of 100 records
0x1FFE0184 slot structure K	Confirmed; it is the link/transmit control block (1, 2.3, 5.12)
0x1FFF9660 four records, stride 260	Confirmed
0x1FFFA0B8 language byte at +0x16	Confirmed; the block is the persistent configuration
0x1FFF9448 remote-control gate word at +0x108	Corrected: 0x1FFF9550 is the general event/transmit flag word (section 8); only bit 2 concerns remote control

Row	Verdict
0x1FFE07E0 charge limit table, stride 0x16	Confirmed
0x1FFFA00C block used by 20, F0, FC	Confirmed; also F2, F4, F6 (update state)

7.8 “0xE0 reply, handler 0xB634”

Statement	Verdict
Reply byte 0 is the constant E1	Confirmed
“Nothing in this function says which transport a type belongs to”	Corrected by the route analysis: type 6 = BLE, type 1 = USB host (section 1)
Type 6 returns 18 bytes: 01 06 00 33, MP305B 00 00, 02 00 02 00, request’s last byte	Confirmed (emu)
Other types return 31 bytes with the listed fields	Confirmed (emu for types 1, 3, 5); no suffix in this layout
The handler reads the word at absolute 0x1C and copies 8 bytes from it + 0x0C	Confirmed
The app’s vector word is 0x0007AFC8, +0x0C gives hardware revision then version; the 8 bytes are not in app.bin	Confirmed. New (inferred): the bootloader’s vector 7 points to its identity block the same way (the update verifier 0x1FE0C relies on vector 7 of an image), so the 8 bytes are the bootloader’s hardware revision and bootloader version
“WebLink’s USB field names do not line up with either layout”	Corrected: the long layout is exactly serial 8 (the model string), hardware 4, bootloader 4, application 4, name 10
Bluetooth captures match the type 6 order with 01 06 00 28	Consistent with the short layout (not re-checked against the captures)

7.9 “0xC6 command, handler 0xCA44”

Statement	Verdict
Reply 0 or FF, length 2 or 3; first failing check stops later stores; output byte not read	Confirmed (emu at every bound)
Check/store table rows 0..11	Confirmed, including <code>screenDirection</code> validated but not stored
Payload 9, 10 not in C4; <code>usbLine</code> at 11; WebLink’s view	Confirmed for the image side
Settings worker 0x128A4 and the normalisation tables	Not re-checked in this pass (settings area)

7.10 “0x18 bind”

Statement	Verdict
S+0x47 = 1, type at K+6, no-reply path	Confirmed (emu)
19 built at 0x2690, called from 0x3836; 0 if S+0x46 != 0 else FF; zero byte for caller type 6	Confirmed (0x12690, from the transmit service 0x133BC; emu)

Statement	Verdict
0x1478C and 0x1483C set/clear S+0x46, S+0x47, bit 1; which front-panel action calls which is open	Resolved: processor 0x2478C and 0x2483C are the allow and deny button callbacks of the bind prompt (buttons 0x1FFE06C0 and 0x1FFE06C4, registered in 0x21A50)
After 0x2690 returns, 0x3836 clears bit 1	Confirmed; it also clears S+0x46 and S+0x47

7.11 “Language, 0xA0 and 0xA2”

Confirmed. Clarification: all A2 side effects (store, output off or charger stop, configuration save, S+0 = 1) happen only when the value changes (emu).

7.12 “0x00, handler 0xDFFC”

Confirmed.

7.13 “Remote-control tail”

Statement	Verdict
C8, E2, E8, EE share one skeleton	Confirmed
Bit 2 skips the body and still builds a reply	Confirmed (deferred-reply mode)
rc must be below 3; mode 0/1/2/3 or FF	Confirmed
rc 2 stores S+0x45; with S+3 == 2 both set; else length 0	Confirmed
rc 0 clears S+0x42	Confirmed, only in the matching mode
rc 1 requires the grant or status 1	Confirmed
E2 and E8 are where Ghidra dropped a block	Confirmed

7.14 “Program table”

Statement	Verdict
Ten slots, ids at +0xAA, per-slot flag at +0xA0, count at +0xB5, selected at +0xB4, 16-byte record per slot	Confirmed. The “flag” is the step count and the 16-byte record is the program name
D4 reply: count, then for ids 1..count the slot’s 16 bytes and flag; variable length	Confirmed; new: ids not contiguous from 1 are omitted
DC	Confirmed
D8: r3 = 0 at 0x2FB0, scan from slot 0, K+4, 0x1FFFA8C4+6, K+5	Confirmed; also K+3 = 0. New: an unknown id produces no D9 at all
Worker 0x1D380 reads 0x4B0 bytes from 0x162000 + slot*0x1000 into 0x1FFF8F7C, sets bit 7; service 0x133BC calls 0x15CA0; D9, id, up to ten 12-byte records, cursor 0x1FFE0187; route suffix for type 6; full chunks 122 bytes; bit 7 cleared at the step count	Confirmed (emu with 3 and 12 steps). New: the destination and suffix come from S+3, not from the route of the D8
D6: id 1..10, find or allocate, increments the count, copies 16 bytes and a flag, two further bytes, second == 1 deletes and renumbers; 0/FF; dispatcher clears +8 and calls 0xAEBC(0)	Confirmed. New: the first of the two bytes is the save flag, the second the operation (0 write, 1 delete, 2 write with steps to follow), see 5.11

Statement	Verdict
DA: id 1..9, looked up, up to ten steps of three words, 30500, 5100 unless the third is 0, third at most 99990	Confirmed (emu at each bound). New: exactly ten records are read every call, an id without a slot is FF, the cursor only resets on D6
5100 = 5.100 A; 30500 reads as 30.500 V if a step is 1 mV	Confirmed: the runner divides the first word by 10 before setting the 10 mV target, and the UI prints it as mV
“99990 is the constant only”	Resolved: the third word is the step duration in seconds
Accepted steps at 0x1FFF8F7C, stride 12; filling the count returns 0	Confirmed; the deferred DB 00 follows the flash write
A rejected value sets FF and 2 at S+0x4B; the dispatcher writes 1 afterwards	Confirmed. New consequence: the busy flag stays 1 until a later successful save
DE table rows 0..64	Confirmed byte for byte (emu). Names and units in 5.15
E2 layout offsets 0..3, mode change guard, action to S+0x4C, enabling needs steps and no faults, output off clears S+0x4C, sets S+6, calls 0xCB8C(1); disabling can call 0xAEBC(0); success sets S+0x49; status 0/1/FF	Confirmed (emu). New: action 1 previous step, 2 pause/resume, 3 next step; S+6 starts the program runner; 0xCB8C is a beep

7.15 “PD profiles”

Statement	Verdict
D0: echoes payload 0, index = byte - 1, class < 0x65 -> 7 groups else 9, 16 bytes, class, count, groups from +0xA0 + index*0x24	Confirmed (emu); no index check
D2: index below 10 or FF, 16 bytes, class, a count, up to 9 groups, groups past the count masked 0xFFF8, dispatcher writes 1 at S+0x4B	Confirmed, with one correction: a save byte follows the count before the groups (D2, id, name[16], class, count, save, groups). New: the dispatcher also requests the output off; without the save byte the busy flag stays set
E4: 0x1FFFA138+0x212 plus 1	Confirmed; also sent unsolicited (bit 10)
E8: offsets 0..5, mode-change path, enable requires no faults, sets selector, S+0x2A, S+0x2C; disable can call 0xAEBC(0); dirty flag	Confirmed (emu). New: S+0x9C and 0x1FFF9BB9 are stored before any check; S+0x9C is a PDO enable mask; the output is switched on only after the PD companion acknowledges (5.20)

7.16 “Charge”

Statement	Verdict
EA returns 0x15/0x16, table	Confirmed; names in 5.21
EC returns 0x1F/0x20, table, payload 26 low half ORed with S+0x9E, 28 high half	Confirmed; names in 5.22
EE payload offsets, type below 6, voltage between table entries 0 and 10, cells limits 7/9/13, type 5 not stored, current below 5001, output bit with S+0x4B	Confirmed (emu at every bound)
Output 1 calls 0xA134 and 0xD6FC when S+0x7B == 0 and S+0x9E == 0; output 0 calls 0xD858 when S+0x7B != 0; both call 0xCB8C(1)	Confirmed, except that the beep only happens when the charger is actually started or stopped (emu)

Statement	Verdict
Mapping each numeric type to a chemistry needs a UI trace	Resolved: 0 LiHv, 1 LiPo, 2 Lilon (Li-ion), 3 LiFe, 4 Pb, 5 NiMH/Cd; for type 5 the u16 is the -dV threshold in mV

7.17 “Connection state and list ingest”

BD, BE, BB statements: confirmed (emu). Additions: BD from type 6 is the CH58x reporting whether a BLE host is bound (BD 01/BD 00); BE selector 0 also copies pending flags to S+0x1D/S+0x1E.

7.18 “0x51, 0x53, 0xF1 and 0xFD”

Confirmed. Additions: the meanings in 2.3 and 5.25; FD 03 also sets 0x1FFF9438 = 1; F1 completes the pre-update handshake.

7.19 “0xE1 received”

Confirmed; the four bytes are the CH58x’s application version (bytes 17..20 of the long E1 layout).

7.20 “Block write and the 0xF0 to 0xFE commands”

Statement	Verdict
20 subcommand 5: offset at 5..8, length at 9..12, data at 0x1D; offset 0 erases 0x100000..0x130000 via 0xBDC6 and clears the checksum; length must be 0x80 and <code>offset + 0x1000 < 0x44000</code> ; counter +0x0C += 0x80; write, read back, add each byte; reply opcode, subcommand, offset, status, length 7	Confirmed (emu)
Subcommand 6 compares the word at 0x0D..0x10; match echoes 6, mismatch 0x86 and clears the counter; bytes 2..5 checksum, byte 6 zero	Confirmed (emu)
Type 6 appends 0x31	Confirmed. New: any other subcommand returns the single byte 31 for type 6
0xBDC6 sends 0xD8 to 0xBF9C between 0xF420(6,4) and 0xF3CC	Confirmed: 64 KB sector erase (06 write enable, D8, 04 write disable)
F0: byte AC, sets 0x1FFFA00C = 1, F1 00	Confirmed
F2: clears 0x98 bytes, first byte 1, 0xFC24, status	Confirmed. New: 0x1FC24 erases block 0 and the target range, and loops 65536 times for a length below 0x10000
F4: 0xFC94, F5 00 + request bytes 2..5 + status	Confirmed. New: it programs 128 bytes and sums the read-back words
F6: 0xFE0C, failure erases 0xF0000	Confirmed. New: success writes the identity magic AA55CC33 into the staged image
FC: byte CA, walks 0x1F0000 in 16 blocks of 0x100, status 1, 3 or 7 from the counter, four records through 0xBF3A, FD 00	Confirmed; the four records are the update header at SPI 0 (length, sum, count, magic), written only when 0x1FFFA010 != 0
FE: AA 55 -> 0xDBE8, 0xCA60(200); 0xDBE8 sets 0x1FFE01CC bytes, calls 0xD868; no reset register; not a bootloader entry	Corrected: FE AA 55 is a factory reset followed by a reboot. 0xCA60(200) starts a countdown in 0x12850, which ends in 0x1B71C: 0x1234 written to 0x2005F000 and AIRCR = 0x05FA0004

7.21 protocol.md section 3, “Opcode matrix”

Statement	Verdict
Response = request + 1, except 20; D8 and 18 answered by deferred D9 and 19	Confirmed. New: C9/E3/E9/EF (remote prompt), DB (after the last DA chunk) are also deferred, and C5, DD, E5, EB can arrive unsolicited (section 8)
Matrix rows E0, 18, C2, C4, C6, C8, D0..EE, A2	Confirmed. Refinements: BD is the CH58x’s link report, not a host command; DE serves modes 1 and 2 as listed; E2 “run/pause/step” is action 1 previous, 2 pause/resume, 3 next plus output start/stop
18 “BLE (AF02)”	The main firmware accepts 18 on any route; the CH58x forwards it from AF02
0x18 sets a flag; 0x12690 emits 19 after a UI decision	Confirmed
The switch also accepts 00 20 51 53 A0 BB BE F0..FE	Confirmed
Unrecognised requests get no reply	Confirmed

7.22 protocol.md 4.1, C2 -> C3

Row	Verdict
Handler 0x58BC, 37 bytes	Confirmed
outState 1 CV, 2 CC	Confirmed; 0 = output off (confirmed in code), 3 = output held above the setpoint (inferred)
batteryState internal enum	Refined: 0 on battery, 1 external input charging, 2 charge held (7.1)
percentage 0..100 %	Confirmed as the battery state-of-charge byte
voltage 10 mV	Confirmed in code (INA226 mV / 10); reads 0 with the output off below 0.501 V
setVoltage 10 mV, current 1 mA, setCurrent 1 mA	Confirmed in code; current reads 0 with the output off
workingTime s	Confirmed in code (calibrated ms counter / 1000)
energy 0.1 Wh	Confirmed in code (mWh / 100, rounded)
power 10 mW	Confirmed in code (μW / 10000, rounded)
currentOver: setting, 0 CC, 1 OCP	Confirmed: 1 trips fault bit 5 after the OCP delay
realChange bit 0 voltage, bit 1 current live updates	Confirmed in code: the bits select live knob editing per field (0x5A270, 0x57064); while the output is on, any nonzero value also sets the UI refresh period to 100 ms instead of 200 ms (0x1E284)
voltageSlow step or ramp at the settings’ ramp rate	Confirmed: ramp of slopeSteps mV per 100 ms, also at switch-on from 0 V
output 0/1	Confirmed; it is the actual power-stage state, not the request
model 0..3	Confirmed
voltageBoard, currentBoard UI keypad input flags	Consistent: persisted UI entry-mode flags, default 1 (inferred)
temperature °C	Refined: signed int8, the companion’s first temperature, which the firmware uses for the battery protections
chargeError bitmask, see 4.5	Confirmed; producer and every bit in 7.2

Row	Verdict
<code>wavePause</code> waveform streaming active flag	Corrected: 1 if the device waveform page timer pointer is zero, otherwise <code>S+0x11</code> . The reply field alone does not identify host streaming
<code>waveTime</code> timestamp in ms	Refined: raw output-on time in ms, the uncalibrated counter behind <code>workingTime</code> , reset by refresh and mode change
<code>outState</code> 0 probably “neither, output off” (inferred)	Confirmed in code

7.23 protocol.md 4.2, C8 -> C9

Statement	Verdict
WebLink sends <code>rc = 2</code> then full <code>rc = 1</code> frames	Consistent with the handler
Payload 11 bytes, field table	Confirmed
<code>refresh</code> 1 = reset cumulative counters	Confirmed: energy and time (<code>0x1A5FC</code>)
<code>model</code> target mode (0 = DC)	Confirmed; 1..3 request a mode change while the output is off
Status 0, 1, FF	Confirmed; the deferred reply uses 0 (allowed) and 1 (denied)
3050 / 5100 limits, FF in other modes, epilogue can change grant state, no rollback, <code>rc > 2</code> rejected, <code>rc = 1</code> needs the grant	Confirmed (emu)

7.24 protocol.md 4.3, C4/C6

Byte layouts and the C6 check table: confirmed (emu). The discrete lists and units are WebLink’s; this pass adds that slope is used as mV per 100 ms by the ramp (`0x19E68`) and the OCP delay as milliseconds (`0x19E68` compares it with a millisecond counter).

7.25 protocol.md 4.4, E0 -> E1

Confirmed, with the corrections above: layout chosen by route (6 BLE short, others long); the long layout matches WebLink’s USB field list; its bytes 8..15 (payload offsets) are inferred to be the bootloader’s hardware revision and version.

7.26 protocol.md 4.5, chargeError

Bit	Verdict
0 <code>OUTPUT_REVERSED</code>	Consistent: ADC1 channel 3 above 0.5 V (sensor meaning inferred)
1 <code>LOW_BATTERY</code>	Confirmed in code
2 <code>BATTERY_LOW_TEMP</code>	Confirmed: below -19 °C
3 <code>BATTERY_OVERHEAT</code>	Confirmed: above 57 °C
4 <code>SYSTEM_OVERHEAT</code>	Confirmed: other sensors at or above 85 °C
5 <code>DC_OUT_OCP</code>	Refined: software OCP after the OCP delay with <code>currentOver = 1</code> , not a hardware trip
6 <code>DC_OUT_OVP</code>	Confirmed: 33.001 V or more for 500 ms
7 <code>DIC_INIT_ERROR</code>	Confirmed: power-stage start-up failure

Bit	Verdict
8 DC_OUT_VOL_FAIL	Confirmed: INA226 and ADC voltage disagree by 5 V or more in CV
9..15 charger faults	Corrected: the reviewed producers set only bits 0..8 in S+0x9E (the C3 field). The separate charger word contributes bits 10 and 11 to EC; indirect writes remain outside this absence claim

7.27 protocol.md 4.6

Statement	Verdict
18/19 behaviour	Confirmed
A0/A2 behaviour	Confirmed
DA limits	Confirmed; the third word is seconds
PD profile is 16 bytes, a class byte and 7 or 9 groups; E4 is index + 1	Confirmed; the 16 bytes are the name, the class is the wattage
EA 20 payload bytes, EC 30, DE 68	Confirmed
20 is a block write replying 20; FE not a bootloader entry	Confirmed for 20; FE AA 55 is corrected to factory reset plus reboot

8 Host-link comparison

Full source: hostlink analysis.

Statements are grouped by source file. C = confirmed, X = corrected, N = new.

8.1 protocol.md 1.1 to 2.2 (framing)

- 1.1 “USB VID 0x28E9, PID 0x028A, report ID 1 OUT / report ID 2 IN, 63 data bytes, 62/61-byte chunking”: **C** (CH58x descriptors and 0x3E1A 62-byte queue; the 61-byte figure is WebLink host-side).
- 1.1 “WebLink de-stuffs AA AA -> AA; firmware doubles AA throughout the framed body, single leading delimiter”: **C** (encoder 0x1FEDC, emu round-trip).
- 2.1 “TX [cnt] AA 12 plen cmd ... cksum, RX addr byte, opcode index 4/1/0”: **C** for the byte order. **N**: the 12/31 etc are (source<<4)|dest nibbles, not a fixed device address; the MCU-side address is computed by nibble swap.
- 2.1 “BLE AF01 replies carry address byte 0x31”: **X (clarified)**. 0x31 is the CH58x AF01 route marker, appended by the CH58x on AF01 writes, echoed by type-6 handlers, and moved to the front by the CH58x reply router 0x3FAC. It is not a bus address and the main MCU never emits it as a frame address.
- 2.2 “checksum = sum of addr, plen, cmd+payload; AA doubled; single leading AA; parser splits first body byte into nibbles, does not compare with 0x12”: **C** (emu, disasm 0x1FF38/0x1FEDC).

8.2 firmware.md “Command dispatcher”

- “0x2F34 watches four flag bytes, first set selects a record at 0x1FFF9660 stride 260, byte 0 type, byte 2 length, byte 4 opcode”: **C** (the four flags are 0x1FFE01AC..AF; addresses given here).
- “two call shapes; 0xBB/0xBD/0xBE no reply buffer”: **C** (disasm of the call sites).
- “type 6 appends one byte copied from the end of the request; 0xF2/0xF4/0xF6/0xFC/0x20 append constant 0x31”: **C**, and **N**: the appended byte is the CH58x route marker (0x31 AF01 / 0x00 AF02), which is why the constant 0x31 is used where the reply always targets AF01.
- “type byte 6 is not a transport name; 0xBD treats 5 and 6 as different”: **C** and explained: type = source nibble; 6 = BLE host, 5 = accessory link, 1 = USB, 3 = CH58x internal.

8.3 firmware.md “Frame parser”

- “Outbound stuffing 0xFFFF0, inbound 0xFF38, AA counter, state table branch, states 1-4 as described, checksum coverage”: **C** (all re-derived by disasm and confirmed by emu round-trip and edge cases).
- **C (offsets, corrected during recovery)**: firmware.md is correct: main state is at +0x208, the fill index at +0x20C, and the AA counter at +0x213. The earlier independent note’s proposed correction was rejected after instruction inspection and stepped execution.
- “0xFF38 is a file offset; absolute 0x1FF38”: **C** (function at 0x1FF38; the wrapper 0x146B8 tail-calls it).

8.4 firmware.md “0x18 bind”

- “0x18 inline 0x3126 sets S+0x47, stores type at K+6, no reply”: **C** (absolute 0x13126).
- “0x19 built at 0x2690 from 0x3836; byte1 = 0 if S+0x46!=0 else 0xFF; len 2 or 3 (type 6 appends 0)”: **C** (absolute 0x12690, emu-confirmed 19 00 00 / 19 FF 00).
- “0x1478C sets S+0x46=1, S+0x47=0, bit 1 of 0x1FFF9448+0x108; 0x1483C clears both, sets same bit; which button calls which is open”: **X**. These are **file offsets**: the real functions are 0x2478C (Allow, sets S+0x46=1) and 0x2483C (Deny, clears S+0x46), both setting bit 1 of 0x1FFF9550. firmware.md’s “0x1483C” should be 0x2483C (= file offset 0x1483C). The button binding is now resolved: 0x2840C builds the confirm-bind dialog and registers 0x2478C on child 0x1FFE06C0 (Allow) and 0x2483C on 0x1FFE06C4 (Deny) via 0x21A50.
- “after 0x2690 returns 0x3836 clears bit 1”: **C**.
- **N**: bind timeout auto-dismisses the dialog and yields 19 FF; the main MCU never stores host IDs (remembering is CH58x-only); USB 0x18 is answered the same way.

8.5 firmware.md “Stream and list ingest” / “Connection state and list ingest”

- “0xBD 0x4660 type 6 stores request byte 1 at S+3, clears K+2 and u16 at K+0x12 when 0; type 5 stores byte 1 at S+4, copies len-2 bytes to S+0x58”: **C** (absolute 0x14660). **N**: S+3 is the host link state (BD 01 up / BD 00 down) and S+0x58 is the accessory name.
- “0xBE 0x855C selectors 0/1/2/4, wheel K+0x2C”: **C** (absolute 0x1855C); **N**: this is Bluetooth-accessory input (wheel + buttons) routed into the LVGL keypad reader; also injectable by a BLE host on AF01.
- “0xBB 0x5AA0 clears 0xC4 bytes at 0x1FFF9A70, count, stride 0x26, sum, sets S+0x3D”: **C** (absolute 0x15AA0); **N**: it is the “Bluetooth Accessories” scan-result list.

8.6 firmware.md “0x51, 0x53, 0xF1 and 0xFD”

- “0x51/0x53 type 3, payload 0, store 0 at 0x1FFF9448+0x103/0x104, load 1000 at 0x1FFF9440”: **C**. **N**: they are the acks of the MCU’s own 0x50/0x52 companion commands.
- “0xF1 inline 0x30D8 type 3, requires 0x1FFFA00C+2 == 1, stores 2”: **C**; **N**: ack of F0 AC update-prepare.
- “0xFD inline 0x32C0, payload 3 or 0xFF; 3 sets 0x1FFF9449, calls 0xAE70(1), ORs 0x2000 (bit 13, request the CH58x identity with E0); 0xFF clears 0x1FFF9449 and bumps 0x1FFF943C”: **C** (absolute 0x132C0); **N**: FD carries CH58x update progress. Note **N**: FD 03 also arrives during boot from node 3 and marks the companion initialised (0x1FFF9449 = 1); a FD 03 from node 1 (USB) is accepted too (emu).

8.7 firmware.md “0xE1 received”

- “0x3C40 acts only when type is 3; if byte 8 of 0x1FFF942C is 0 sets K+0=1 and copies 4 bytes from record+0x11 to bytes 8..11, else to 0..3 and 4..7; meaning not named”: **C** (absolute 0x13C40). **X/N**: the 4 bytes are the **CH58x version/identity field** (01 00 00 0f on the emulated reply), copied to 0x1FFF9434..37 then 0x1FFF942C..33, and shown by the sys-info UI 0x57D54. It is not a BLE MAC (the MAC stays inside the CH58x).

8.8 protocol.md 1.3 (bind table)

- “device shows allow/deny prompt every connection; reply on button press”: **C** (prompt 0x2840C, callbacks 0x2478C/0x2483C).
- “19 00 allowed, 19 FF denied”: **C** (emu).
- “device does not remember an allowed host with WebLink’s frame”: **C** in code: the main MCU stores nothing, and the CH58x only stores a host ID after a 19 00 with the frame’s last byte zeroed (CH58x side); WebLink’s frame path goes through the prompt, not the remember path.
- “after deny, 0xC4/0xC2/0xE0 still answered; binding does not protect reading”: **C** (dispatcher has no bind gate; emu `test_dispatch` answers reads regardless of S+0x46).
- “how long the device waits before giving up, and what it replies”: **answered (code)**: the popup auto-dismisses (tens of seconds) and the deferred 0x12690 then replies 19 FF.
- “whether USB shows the prompt”: **answered (code)**: USB 0x18 is dispatched and answered identically; the prompt UI would appear, but USB control is gated by the remote grant (granted without a prompt), not by bind.

8.9 firmware-architecture.md “Framing and bridge routing”

- “decoded slot: source nibble, dest nibble, length, unused, opcode, payload; addr = (source«4)|(dest&15); wire = AA addr len opcode payload cksum; AA doubled incl addr/len/cksum; leading AA single; addr 0xAA cannot be received; zero length fails; checksum change rejected”: **C** (all re-derived independently by disasm + emu here).
- “type-6 suffix is an internal bridge route byte, AF01=31 AF02=00, removed before notification; update handlers use fixed 31”: **C** (CH58x 0x4074/0x3FAC, and MCU type-6 append; emu both sides).
- encoder 0x15430, body 0x1FEDC, stuffing 0x1FFF0, decoder wrapper 0x146B8: **C** (all confirmed).

8.10 hardware-buses.md “Main processor UARTs”

- “0x4001CC00 init 0x644A4 115200, RX 0x1EF0C decoder ch0 parser 0x11900, P3.9/P3.8 mux 0x20/0x21 -> PD companion”: **C**.
- “0x4001D400 init 0x1787C 115200, RX 0x1EF5C decoder ch1, P2.0/P2.1 mux 0x20/0x21 -> BLE/USB bridge, four mailboxes stride 260 at 0x1FFF9660, sources 6/1/5/3”: **C** (this is the hostlink UART).
- “0x40021000 1000000 baud, RX 0x1F048 text-line buffer, CR terminator, buffer 0x1FFF8A8C 512 bytes, status/len/cursor 0x1FFE0148/14A/14C, console/factory”: **C**, and **N**: full command set documented (section 1); consumed by 0x1227C in `Time_task`; not reachable from the host link.
- “source 3 opcode 0x55 handshake path and opcode 0x11 reaches 0x13C88”: **C** (0x55 ack in 0x133BC; 0x11 -> 0x13C88 time sync in the RX callback fast path).
- “DMA base 0x40053400 channels 0/1; selectors 0x12C/D/F and 0x14C/D/F; NVIC priority arg 15”: **C**.
- **N**: RX is per-byte interrupt (RI), DMA is transmit only.

8.11 firmware-permissions.md / firmware.md remote-control

- “bind decision S+0x46 and remote grant S+0x42 are separate; decline callback clears S+0x46/S+0x47 but not S+0x42”: **C**.
- “reads answered after denial”: **C**.
- “C8 remoteCon 2 grants immediately for connection type 2, pends for type 1/0”: **C** and refined: **type 2 = USB** (grants at once, no prompt); **type 1 = Bluetooth host** (pends for the Allow dialog); type 0 = no link (dropped). emu-confirmed.
- **N (answers TBD-005/TBD-012)**: on link drop S+3 -> 0, the UI pass 0x1E284 clears the remote grant S+0x42; the output enable 0x1FFFAA2E is not touched, so the output stays on. There is a keepalive (0x55) and a retransmit-then-give-up watchdog (about 50 tries / 5 s) plus an 8 s USB-silence timeout that clear S+3.

8.12 New, not in docs/research

- The complete 0x1FFF9550 work-mask bit map (section 6).
- The node numbering (1/2/3/5/6) and its exact meaning per source.
- The internal MCU<->CH58x message table with E0/FC/50/52/10/F0/55 outbound and 55/51/53/F1/FD/E1/11/BD/BB/BE inbound, all emu-verified.
- The USART7 console command set and that it can switch the output on with no bind/grant.
- The E1-received 4 bytes identified as the CH58x version field.
- The transmit service 0x133BC as the single point that serialises all outbound frames, with one outstanding frame gated by K+0.

9 CH58x comparison

Full source: ch58x analysis.

9.1 protocol.md 1.1 USB HID

Statement	Result
VID 0x28E9, PID 0x028A, bcdDevice 2.00, string indexes 1, 2, 3	Confirmed [A]. New: index 3 (serial) is not served; the request stalls
Strings wch.cn and MP305B	Confirmed. New: the product string is RAM and is replaced by the 8-byte name from DataFlash 0x6E0E (cut at the first space) when USB is enabled
WebLink filters on usagePage 1, usage 4	The descriptor's top-level usage is 0 (09 00). With this image the WebLink filter would not match. Needs a USB check (TBD-013)
WebLink checks the product name contains MP305B/MP305A/MP3010B	Consistent; the name comes from the MCU's FC (MP305B on 1.6.0.51)
EP1 OUT/IN interrupt 64 bytes 1 ms, HID 1.11, report descriptor 35 bytes, 100 mA	Confirmed [A]. New: attributes 0xC0 (self powered)
Report ID 1 OUT 63 bytes, report ID 2 IN 63 bytes	Confirmed [A]
Outgoing chunking, stride 61, first report [60, F[1..60]]	Compatible, not required: the firmware accepts any count 1..62 per report and does not check the report ID
Frames of 63 bytes or fewer in one report	Corrected detail: at most 62 stream bytes fit in one report after the count byte
5 ms delay after each report	Not needed by the CH58x (it reassembles across reports); but a new frame must not start before the previous one was forwarded (about 1.25 ms), and frames must not share a report [E]
Firmware doubles 0xAA in the framed body; de-stuff in pairs	Confirmed for USB IN [E]. New: the CH58x also de-stuffs and checks the checksum of OUT frames, then re-encodes them; bad frames are dropped silently
Multi-report replies arrive with a leading chunk count byte	Corrected: the byte after the report ID is the number of valid stream bytes in that report (1..62), in every report
WebLink does not verify checksum or address	WebLink behaviour; the CH58x has already verified the checksum of every IN frame it forwards
A USB connection has not been observed	Still true; all USB statements here are [D]/[E]

9.2 protocol.md 1.2 Bluetooth LE

Statement	Result
Local name prefix 0000MP30	Confirmed and explained: template 0030MP305B, index 2 patched to '0' every second unless RAM 0x200043B7 == 0xA5 (no writer setting it to 0xA5 was identified)
WebLink trims the first four characters	WebLink behaviour; the four characters are constant 0000
Service UUID AF00 in the advert	Confirmed: incomplete 16-bit UUID list 03 02 00 AF
Manufacturer data company 0xABBA, prefix affa0135	Confirmed. New: AF FA fixed; 01 35 02 00 set by the MCU's FC; last 14 bytes set by AF01 10/C0 and persistent
AF01 read, write, notify: commands and telemetry	Confirmed properties 0x1A. New: reads return 20 zero bytes; first byte of every write is discarded
AF02 read, write, notify: bind and hardware info	Confirmed properties. Corrected: E0 on AF02 is forwarded to the main MCU like any other opcode; AF02 has two local opcodes: 00 (link status and BD address) and 18 with fast flag
FEE0/FEE1 bootloader OTA service only	Consistent: not present in this application image
GATT table: 180A 2A23..2A2A and 2A50, AF00, DB00/DB01	Confirmed [D]; handle spacing in the capture fits the registration order 180A, AF00, DB00
DB00 purpose unknown	New: DB01 does nothing (empty hooks, never notifies)
180A holds vendor placeholders	Confirmed: exact values listed above
Negotiated MTU 247 on macOS	Consistent: 247 is the maximum this configuration allows (BufMaxLen 251)
Device not seen while connected (inferred)	Confirmed in code: one host link, advertising re-enabled only on disconnect, a second central is dropped
Advertisements several seconds apart; scan at least 10 s	Advertising interval is 500 ms [D]; advertising is off until the MCU enables it after boot, while a host is connected, while a USB host is active, and when remote is disabled. The long gaps seen on macOS are not explained by the interval [I: scanner duty cycle]
Notifications must be enabled on AF01 and AF02 before commands	Confirmed in effect: a reply to a characteristic whose CCCD is off is dropped and the MCU retransmits it for about 5 s. CCCDs reset on every disconnect
BLE frames unpadded, un-stuffed, no length or checksum	Confirmed: the CH58x adds and removes the UART framing
Single writes/notifications without chunking	Confirmed: no fragmentation in either direction
MTU >= 126 needed for full-size frames	Not a CH58x limit; the need follows from the longest MCU reply plus 3 (plus 1 on AF01)
Replies on AF01 carry the address byte 0x31	Corrected: 0x31 is the route tag the CH58x appended to the request and moves to the front; it is constant, not an address
Writes with response work on AF01 and AF02	Consistent; write-without-response is not declared
Round trip 120 to 135 ms	The CH58x adds about 1.25 ms per direction plus UART time (a 41-byte frame takes 3.6 ms at 115200) [I]; most of the time is elsewhere

9.3 protocol.md 1.3 Bind handshake

Statement	Result
WebLink writes the bind frame to AF02 after enabling notifications	Consistent
Field boundaries (class byte, 16-byte ID, fastBinding, status) inferred	Corrected by firmware: the CH58x takes bytes 1..16 as the ID and the last byte of the write as the fast flag; other bytes are ignored. In WebLink's frame the ID is 00 08*14 00 and the fast flag is 0
Prompt at every connection; reply after a button press	Consistent: fast flag 0 always forwards to the MCU
19 00 allowed, 19 FF denied	Confirmed; the fast path produces the same two replies
Device does not remember an allowed host with the WebLink frame	Explained: after 19 00 the ID is stored, but with fast flag 0 the stored IDs are never consulted
After a deny, reads are still answered	Consistent: the CH58x forwards all writes regardless of bind state. New: an unbound link is terminated about 30 s after connecting
Whether fastBinding = 1 lets the device remember a host	Firmware: yes, if the same 16 bytes were allowed once before (up to 5 IDs); an unknown ID with fast flag set gets 19 FF without a prompt
Whether USB shows the prompt	The CH58x has no bind logic for USB; any USB behaviour is the MCU's
How long the device waits for a button press	CH58x side: the link is dropped 30 s after connection unless bound; the MCU's own prompt timeout is not in this image

9.4 protocol.md 2 Frame encoding

Statement	Result
HID TX [cnt] AA 12 plen cmd payload cksum	Confirmed (report ID 1 precedes cnt on the wire)
HID RX [cnt] AA addr plen cmd payload cksum	Confirmed; addr is 0x21 [M]
BLE AF01 TX 12 cmd payload	Confirmed; the 12 is discarded, any value works
BLE AF01 RX addr cmd payload	Corrected: the first byte is the route tag 31
BLE AF02 TX/RX without address	Confirmed
Checksum 8-bit additive; stuffing on HID only	Confirmed
Checksum = sum of bytes 2..N-1	Confirmed (address + length + data) [E]
If the checksum equals 0xAA another 0xAA is appended	Confirmed [A: 0x35A8]
Target address 0x12 is the device address	Corrected: 0x12 means source 1 (USB), destination 2 (MCU); the MCU replies to the source nibble

9.5 protocol.md 3 and 4 (related entries)

Statement	Result
0xBD “RealTime_DATA”: connection state and peer metadata	Confirmed and explained: BD 00/BD 01 from the CH58x on channel 6 (bind state); BD k name on channel 5 (accessory)
E0 type 6: 17 bytes, final byte is the route suffix removed before notification	Confirmed [E]: e1 ... 00 on AF02, 31 e1 ... on AF01
Which chip each E1 version belongs to is inferred	Resolved: both host-visible versions come from the main image (application 1.6.0.x, hardware 2.0.2.0). The CH58x version is 1.0.0.15 and is only reported to the MCU
No release on disconnect; what happens is unknown	New: after a bound host disconnects the CH58x sends BD 00; the MCU sets connection type 0 (and clears state per firmware.md). An unbound host disconnecting produces no message

9.6 firmware.md “USB device” and firmware-architecture.md USB/GATT/binding

Statement	Result
Report descriptor at data.bin + 0x12F4C, configuration +0x12F70, device +0x12F9C	Confirmed (processor 0x8F4C, 0x8F70, 0x8F9C)
Device and configuration fields	Confirmed
Report descriptor content	Confirmed; exact bytes given above
Queue service 0x3E1A limits chunks to 62; 0x4ECA sends report ID 2, count, bytes	Confirmed [E]
Receive 0x4E86 takes the count from byte 1 and feeds from byte 2	Confirmed [E]; the count is not bounded
AF/DB characteristics properties 0x1A, handles assigned at runtime	Confirmed
Initialized name 0030MP305B, prefix rewrite unresolved	Resolved (see 1.2)
AF02 write 0x40DC, AF01 write 0x4224, forwarder 0x4074 appends 31/00, router 0x3FAC	Confirmed [E]. Addition: the AF01 first byte is dropped in the callback 0x72A6, before 0x4224
Nonzero final byte selects a lookup; reply 19 00/19 FF	Confirmed [E]
Save only when guard 0x20002F88 is zero	Confirmed; the guard is “last notification failed”
Five 16-byte IDs at 0x6F00, oldest shifted out	Confirmed [E]

9.7 firmware.md “Companion image strings”

Statement	Result
CH58x_BLE_LIB_V1.8 at slice offset 0x7E68	Confirmed (processor 0x8E68); the image refuses to run with another library version
0030MP305B at slice offset 0x8572	Confirmed (.data source of 0x20002E9A); it is a template, rewritten at run time
Load address 0x1000	Confirmed

9.8 firmware.md other sections that describe the CH58x's messages

Statement	Result
0x51, 0x53, 0xF1 act only for type 3 and request byte 1 = 0	Consistent: the CH58x replies 51 00, 53 00, F1 00
0xFD with byte 3 or 0xFF	This CH58x only ever sends FD 03
0xE1 received (type 3) copies four bytes from offset 0x11	Confirmed: they are the CH58x version 01 00 00 0F
0xBB records of stride 0x26 (6 + tail)	Consistent with BB list: 6-byte address, length, name up to 31
0xBE: byte 1 = 1, 2, 4; byte at length-2 = 1 or 0xFF	Confirmed: button bits and wheel byte of an accessory report (9-byte frame, wheel at offset 7)

9.9 Captures

All four captures (2026-09-29T193614, 194219, 194319, 201428) re-decoded with the firmware model:

Item	Decoding	Agrees
Name 0000MP305B S E!K	0000 + FC name MP305B + two FC padding spaces + S (remote enabled, BLE not blocked) + template spaces + BD-address suffix E!K ($v \bmod 94 = 36$, $(v/94) \bmod 94 = 0$, $(v/8836) \bmod 94 = 42$)	Yes. The spaces at 10..11 prove the DataFlash name override was present
Manufacturer data affa0135 02 00 + 14 zero bytes (LOGBOOK)	fixed AF FA, MCU FC bytes 01 35 02 00, empty host tail	Yes
GATT handles 15..31, 34, 37, 41	180A, AF00 (3 attributes per characteristic), DB00	Yes
MTU 247	library maximum for BufMaxLen 251	Yes
AF02 TX 18 00 08*14 00 00 00	ID 00 08*14 00, fast flag 0: forwarded as AA 62 14 18 ... 00 FE, prompt	Yes
AF02 RX 19 00 (3.0 to 7.7 s later)	MCU reply 19 00 00, tag stripped; ID saved to 0x6F00; BD 01 to MCU; MCU then detaches USB (50 01)	Yes
AF02 RX 19 ff (194319)	deny; nothing saved; bind state stays 0, so the link would be dropped 30 s after connecting; the capture ends before that	Yes
AF02 TX e0, RX e1 01 06 00 28 ... 02 00 02 00 (17 bytes)	forwarded with tag 00; MCU type-6 layout; tag stripped	Yes
AF01 TX 12 c4, RX 31 c5 5a 02 00 00 01 f4 01 32 00 00 00	12 dropped, c4 31 forwarded; reply routed by tag 31	Yes
AF01 TX 12 c2, RX 38 bytes starting 31 c3	same path; 37 bytes from the MCU plus the tag	Yes
AF01 TX 12 e0, RX 31 e1 ... (18 bytes)	same path	Yes
Reads answered after deny	CH58x forwards regardless of bind	Yes